The evidence, from selection to the shuffling test

1. Discovery

Search-selected rule; the large initial demonstration used a localised random patch.

2. Confirmation

Frozen seeds reproduced irregular, nonmonotone large-domain growth.

3. Composition

Dense regions are mixed excitations; sparse regions are near-empty.

4. Controls

Pure rotation and the active Q+M sibling did not reproduce the same regime.

5. Homogeneous starts

Separate whole-world random ensembles established density dependence.

6. First passage

A geometrical endpoint was followed through longer observation windows.

7. Prepared shapes

Large shapes persisted; relative-growth counts are not absolute formation tests.

8. Intervention audit

Shuffling destroyed measured structure, yet the shuffled states crossed the geometrical endpoint.

9. Intervention result

Shuffling invalidates the first-passage assay for a causal memory effect. The observed reversible domains remain.